Assignment 01: Conditional Statements & Loops

Time: 2 hours

Instructions

- Write your code in .php files, one per question (e.g., q1.php, q2.php, etc.)
- Test each script in a browser or via CLI (php filename.php) before submitting
- Add comments explaining your logic
- Use meaningful variable names

Section A

Q1)

A power company charges customers based on units consumed per month, using slab rates,

- 0–100 units: Rs. 5/unit
- 101–200 units: Rs. 7/unit
- 201–300 units: Rs. 10/unit
- Above 300 units: Rs. 15/unit

Additionally, if the total bill exceeds Rs. 2000, apply a 5% surcharge.

Write a PHP script that takes $units consumed as input and prints the total bill amount, showing the slab breakdown.

Q2)

A bank approves a personal loan based on the following rules,

- Applicant's age must be between 21 and 60
- Monthly income must be at least Rs. 30,000
- Credit score must be 650 or above

If all conditions are met, print "Loan Approved". If age or income fails, print the specific reason. If credit score alone fails, print "Loan Approved with higher interest rate" instead of outright rejection.

Q3)

A cinema prices tickets based on,

- Age below 12 or above 60: Rs. 150 (concession)
- Age 12–60: Rs. 300 (regular)
- If the show is between 9 PM and 6 AM (24-hour format), add a Rs. 50 late-night surcharge
- If the day is Wednesday, apply a 20% discount on the final price (Wednesday offer)

Take $age, $showHour (0–23), and $day as inputs and calculate the final ticket price.

Q4)

A university awards grades not just on marks but also attendance,

- Marks ≥ 90 and Attendance ≥ 75%: Grade A
- Marks 75–89 and Attendance ≥ 75%: Grade B
- Marks 60–74 and Attendance ≥ 75%: Grade C
- Any marks, but Attendance < 75%: "Detained – Insufficient Attendance" (regardless of marks)
- Marks below 60 (with sufficient attendance): Grade F

Use if-elseif-else (not switch) to implement this.

Section B

Q5)

A person invests Rs. 50,000 at an annual interest rate of 8%, compounded yearly. Using a for loop, print a year-by-year breakdown of the investment value for 10 years, and state in which year the investment crosses Rs. 1,00,000.

Q6)

An ATM has denominations of Rs. 2000, 500, 200, 100, and 50 notes only (unlimited supply of each). Using a while loop, write a script that takes a withdrawal amount and calculates the minimum number of notes needed for each denomination. The amount must be a multiple of 50; otherwise, print an error message using a validation if check.

Q7)

A bus has 40 seats. You're given a list (array) of 40 values representing seat status (1 = booked, 0 = available). Using a foreach loop:

- Count and display the total number of available seats
- Display the seat numbers that are available
- If available seats are less than 5, print "Hurry! Few seats left"

Q8)

Simulate a login system where a user has 3 attempts to enter the correct password (hardcode the correct password in your script, e.g., "Secure@123"). Use a do-while loop that:

- Asks for password input (simulate with an array of attempts like ["pass1", "Secure@123"] if not using real-time input)
- Prints "Access Granted" if correct
- Decrements attempts on failure and shows remaining attempts
- Prints "Account Locked" after 3 failed attempts

Section C

Q9)

A restaurant's menu (as an associative array) has items with prices as follows,

```
$menu = [
  "Burger" => 350,
  "Pizza" => 650,
  "Pasta" => 450,
  "Coke" => 80,
  "Ice Cream" => 150
];
```

A customer orders multiple items with quantities (as an array). Using a loop:

- Calculate the subtotal for all ordered items
- Add 5% GST
- If the customer is a "loyalty member" AND the subtotal (before tax) exceeds Rs. 1000, apply a 10% discount on the subtotal before adding tax
- Print an itemized bill followed by the final amount

Q10)

A warehouse wants to visually arrange box counts on shelves in a pyramid layout, where shelf 1 has 1 box, shelf 2 has 2 boxes, and so on up to $n shelves (take $n from user input). Use nested for loops to:

- Print the pyramid pattern using * for boxes
- Calculate and print the total number of boxes across all shelves
- Identify and print which shelf number holds exactly 10 boxes (if any), using a break statement to stop the loop once found

Example for $n = 4:

```
*
* *
* * *
* * * *
```